7 (a) (i) e.m.f. = energy / charge C1

= $(1.6 \times 10^5) / (1.8 \times 10^4)$

= 8.9 V A1

(ii) current = $\Delta Q / \Delta t$ C1

= $(1.80 \times 10^4) / (1.3 \times 10^5)$

= 0.14 A .... A1 [4]

(b) (i) energy $\propto R$ (or formula) C1

energy = $(15 / 45) \times 1.14 \times 10^5$ C1

= 3.7×10^4 J A1

(ii) energy dissipated in internal resistance (of battery) .... B1 [4]

OR in <u>extra</u> resistance in circuit